Compact Radial Kernel Banks for Cancer Classification with Limited Training Data

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Abstract

Training a medical-image classifier with few annotated cases motivates restricting the number of parameters estimated from those cases. We study a compact classifier for this setting, using convolution kernels whose weights depend only on distance from the kernel center. A one-dimensional radial profile specifies each kernel. We jointly train 16 profiles, summarize each convolution response map by three statistics, and fit a logistic classifier to these statistics. We then cluster kernels with correlated training-image responses and replace each cluster by one averaged profile. On the breast dataset, this procedure reduces the bank from 16 to nine kernels while retaining test area under the receiver operating characteristic curve (AUC): 0.9415 versus 0.9411. A bank of 90 predefined kernels reaches 0.9387, and a convolutional neural network reaches 0.9626. On the brain dataset, the same reduction lowers test AUC from 0.7624 to 0.7335, despite meeting the validation criterion. Thus, fewer kernels preserve discrimination on one dataset but not the other. An example links a learned profile to the classifier coefficients to show how its contribution can be interpreted. These results support further study of radial kernels for small training sets; they do not establish a training-data advantage, because training-set size was not varied.

1 Introduction

Medical-image classifiers can support cancer detection by identifying image regions that resemble annotated lesions. Training such classifiers is difficult when few annotated cases are available, as may occur for uncommon conditions or specialized clinical cohorts. The relevant constraint is the number of independent cases: extracting many regions from the same image increases the number of training examples but does not provide equally many independent observations. Our goal is to develop a classifier that can be trained under this constraint, including situations in which a larger dataset cannot be obtained.

Convolutional neural networks (CNNs) learn image features by optimizing successive layers of convolution kernels [7, 8]. A smaller set of trainable parameters offers an alternative when the available cases may be insufficient to estimate a large model reliably. This motivates a constrained model, although model size alone does not establish how many cases it needs. That question requires performance measurements across training-set sizes.

Earlier image-analysis methods used explicit descriptions of edges, texture, and local intensity relationships [1, 6]. Gaussian derivatives and difference-of-Gaussians describe intensity changes at different scales [2]; Gabor filters describe oriented texture [3]; and histograms of oriented gradients and local binary patterns summarize local gradients and intensity relationships [4, 5]. Related descriptors have been used in breast histopathology, mammography, cervical histopathology, and thyroid cytology [12, 13, 14, 15, 16, 17]. Other approaches combine texture and deep features [18] or use logistic regression and support vector machines for classification [19]. These studies motivate retaining explicit image measurements while adapting the representation to the available data.

We investigate a *learned radial bank*: a set of convolution kernels, each represented by a one-dimensional function of distance from its center. Training adjusts these radial profiles jointly. After training, kernels with similar responses are grouped and averaged to form a *merged radial bank*. For comparison, a *predefined-kernel bank* selects kernels from specified mathematical families without optimizing their individual weights. These names distinguish profile training, subsequent merging, and selection from predefined candidates throughout the paper.

We evaluate how much classification performance a compact radial model retains on the available breast and brain datasets, as an initial step toward this small-data goal. Interpretability is a secondary objective: the profile describes the spatial intensity pattern measured by a kernel, and the final classifier coefficients describe how its measurements affect the prediction. We provide a worked example of this interpretation. The experiments compare model representations at fixed training-set sizes; the discussion separates what these comparisons establish from the additional evidence needed to demonstrate reliable training with fewer independent cases.

2 Methods

2.1 Problem formulation

An *image region* is a square crop of a source image. Region extraction consists of randomly ordering eligible center pixels, taking crops around those centers, and retaining

crops that satisfy the annotation-coverage and tissue criteria in Section 2.2. Let $x_i \in \mathbb{R}^{H \times W}$ be a grayscale region of height H and width W, and let γ_i be the fraction of its pixels inside a binary lesion mask. We assign $y_i = 1$ when γ_i meets the positive-coverage threshold and $y_i = 0$ when it meets the negative-coverage threshold. Regions between the thresholds are excluded. The training set is $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, where N counts extracted regions, not patients. The task is to estimate a region's probability of belonging to the positive class, $p(y = 1 \mid x)$.

For the kernel-based models, a bank $\{h_j\}_{j=1}^{K}$ contains K convolution kernels. The response map $z_j = x * h_j$ records the result of applying kernel h_j at each image position; $*$ denotes two-dimensional convolution with zero padding. A *pooled statistic* reduces a response map to one number. Concatenating the statistics from all kernels gives the feature vector $\phi(x)$ used by the classifier. Thus, z_j is a spatial map, whereas $\phi(x)$ is a vector of region-level measurements. The learned and merged radial banks use three statistics per kernel; the predefined-kernel banks use one. The CNN comparison learns a multilayer representation separately.

2.2 Data and preprocessing

We refer to the two datasets as *Breast* and *Brain*. Region labels follow the coverage definition in Section 2.1. Breast contains high-resolution TIFF images of approximately 6000×4750 pixels: 127 malignant images, 365 normal images, and 125 annotation files. Malignant-area contours are marked in red. Thresholding the red channel while suppressing green and blue, followed by dilation, erosion, closing, and hole filling, converts these contours into binary lesion masks. We extract 128×128-pixel regions at native resolution. Positive regions require $\gamma_i \geq 0.80$ and negative regions $\gamma_i \leq 0.02$. Negative regions include tissue near and far from the lesion and regions from normal images. Crops with fewer than 35% nonzero intensity values are excluded as background.

Brain is the lower-grade glioma magnetic resonance imaging (MRI) dataset of Buda et al. [11], derived from The Cancer Genome Atlas and The Cancer Imaging Archive. It contains fluid-attenuated inversion recovery images and manual tumor masks from 110 patients. We use the saved 64×64-pixel region dataset. Its preprocessing record specifies $\gamma_i \geq 0.20$ for positive regions and $\gamma_i = 0$ for negative regions. A quarter of the negative-region extraction target is assigned to a band obtained by dilating the tumor mask by 12 pixels, providing examples near the tumor boundary. Background exclusion and per-class caps limit the number of accepted regions. These labels describe tumor coverage within a crop, not a patient-level cancer diagnosis.

The nominal allocation is 70% for training, 15% for validation, and 15% for testing, fixed before model fitting. Breast groups crops by source-image identifier inferred from filenames. Brain groups slices by patient identifier. No source image is intended to contribute to more than one split, and Brain also separates patients. Patient-level separation cannot be established for Breast from source-image identifiers alone. Table 1 gives the region counts; these counts should not be interpreted as numbers of independent cases.

Table 1: Image-region counts for the radial-bank experiments. Breast is split by source image; Brain is split by patient. Pos. and Neg. denote positive and negative regions.

Cohort	Split	Pos.	Neg.	Total
Breast	Train	2,900	3,430	6,330
	Validation	1,080	790	1,870
	Test	1,040	830	1,870
	Total	5,020	5,050	10,070
Brain	Train	3,000	3,000	6,000
	Validation	1,000	1,000	2,000
	Test	1,000	1,000	2,000
	Total	5,000	5,000	10,000

Kernel-based models use grayscale regions. Breast retains the original intensity scale. Brain subtracts each region's mean intensity and divides by its standard deviation to reduce differences in intensity scale. These are the preprocessing choices used in the reported experiments. Section 3.4 evaluates the effect of standardizing Breast; a corresponding comparison without standardization was not reported for Brain.

2.3 Learned radial bank

Radial representation. For an odd kernel width s, let $o = (s-1)/2$ be the center coordinate. A pixel at (a, b) belongs to radial bin $r = \lfloor \sqrt{(a-o)^2 + (b-o)^2} \rfloor$. Let $B_r(a, b) = 1$ for pixels in bin r and zero elsewhere, and let R be the largest bin index in the square kernel. Kernel j is defined by

$$h_j = \sum_{r=0}^{R} f_j(r) B_r. \tag{1}$$

The *radial profile* $f_j(r)$ is the common kernel weight assigned to all pixels in bin r; it is not an image intensity. Equation (1) replaces s^2 separate weights by $R + 1$ radial coefficients. Radial binning ties weights at similar distances and excludes independently oriented patterns. On the square pixel grid, right-angle rotations preserve the kernel exactly; arbitrary rotations also involve interpolation and boundary effects.

Only coefficients with $r \leq R_{\max}$ are optimized. The remaining coefficients keep their initial values. In the saved Breast model the outer bin at $r = 21$ is nonzero, whereas the Brain coefficients beyond $r = 30$ are zero. Thus, $R_{\max}$ bounds the trainable radius, not necessarily the nonzero support of the kernel.

Response statistics. Write $P = HW$ for the number of positions in a response map. Let $T_m(z)$ be the mean of the m largest values in map z, with the integer count $m = \max\{1, \text{round}(\alpha P)\}$. The fraction α is 0.01 for Breast

and 0.10 for Brain. For each kernel we define the *mean magnitude* a_j, *maximum magnitude* b_j, and *signed extreme response* c_j:

$$\begin{aligned} a_j(x) &= P^{-1} \sum_{u,v} |z_j(u,v)|, \\ b_j(x) &= \max_{u,v} |z_j(u,v)|, \\ c_j(x) &= T_m(z_j) - T_m(-z_j). \end{aligned} \tag{2}$$

Here u and v index rows and columns of the response map. The mean magnitude summarizes response strength across the region, the maximum magnitude records the strongest response, and the signed extreme response retains information about response polarity. A localized response can have a large maximum but a small mean. Reversing a kernel's sign leaves a_j and b_j unchanged and reverses c_j. These properties of Eq. (2) help interpret the features but do not assign them a particular tissue meaning without examining the kernel and images.

The feature vector has $3K$ entries, ordered as

$$\phi(x) = (a_1, \ldots, a_K, b_1, \ldots, b_K, c_1, \ldots, c_K).$$

Each entry is standardized using its training-set mean μ_ℓ and standard deviation σ_ℓ. The final logistic classifier is

$$\begin{aligned} \eta(x) &= \beta_0 + \sum_{\ell=1}^{3K} \beta_\ell \frac{\phi_\ell(x) - \mu_\ell}{\sigma_\ell}, \\ p(y = 1 \mid x) &= \big(1 + \exp[-\eta(x)]\big)^{-1}. \end{aligned} \tag{3}$$

A zero or negligible feature standard deviation is replaced by one. Equation (3) makes the contribution of each standardized feature explicit. Logistic fitting provides probability scores; their calibration was not evaluated.

Initialization and training. We initialize $K = 16$ radial profiles and then optimize their coefficients jointly. For Breast, eight initial profiles come from a predefined-kernel composite and its perturbed variants. The composite is averaged over rotations at 10° intervals, after which its weights are averaged within each radial bin to obtain an initial profile for Eq. (1). Composite construction is defined in Section 2.5. The other eight profiles use positive central regions and negative surrounding rings at different radii, with small random perturbations. The random-number seed is 42. Brain uses perturbed variants of a smooth positive-center, negative-surround initial profile. Initialization specifies the starting coefficients; training subsequently changes them.

During profile training, the standardized features in Eq. (2) feed a neural classifier with 32 hidden units, a rectified linear activation, dropout probability 0.1, and one output. Dropout randomly omits hidden activations during training. Feature standardization at this stage uses training responses of the initial bank and remains fixed during optimization. The loss is

$$\begin{aligned} \mathcal{L} = \mathcal{L}_{\text{cls}} &+ \lambda_s \mathcal{L}_{\text{smooth}} + \lambda_c \mathcal{L}_{\text{curv}} \\ &+ \lambda_a \mathcal{L}_{\text{anchor}} + \lambda_n \mathcal{L}_{\text{scale}} + \lambda_d \mathcal{L}_{\text{div}}. \end{aligned} \tag{4}$$

The classification term is binary cross-entropy with the positive-class loss multiplied by N_-/N_+, where N_- and N_+ count negative and positive training regions. The smoothness and curvature terms are mean squared first and second differences of the full profiles. The anchor term is the mean squared change of trainable coefficients from their initial values. The scale term is the mean squared difference between each full profile's Euclidean norm and the reference initial-profile norm. The diversity term is the mean of $\exp(-\|\mathbf{f}_i - \mathbf{f}_j\|_2^2)$ over distinct pairs, where $\mathbf{f}_j$ contains the trainable coefficients. All five regularization coefficients in Eq. (4) are 10^{-4}; these fixed settings were not separately optimized in the reported comparison.

Adam optimization updates the radial profiles and the temporary neural classifier in batches of 128 regions. The classifier learning rate is 3×10^{-3}. Breast uses 31×31 kernels, $R_{\max} = 20$, 25 epochs, and profile learning rate 2×10^{-3}, selected by validation AUC from $\{10^{-2}, 2\times10^{-3}, 5\times10^{-4}\}$. Brain uses 63×63 kernels, $R_{\max} = 30$, 40 epochs, and profile learning rate 10^{-2}. These configurations permit 21 trainable radial coefficients per Breast kernel and 31 per Brain kernel; the Brain kernel spans almost the full region width. Kernel widths were not compared systematically across datasets.

We retain the profiles from the epoch with the highest validation AUC. Their training-set feature means and standard deviations are then recomputed, and the temporary neural classifier is replaced by class-balanced logistic regression on the frozen features, giving Eq. (3). Thus the deployed 16-kernel classifier has 48 feature coefficients and an intercept, although profile training uses an additional neural classifier. This distinction matters when describing model size.

2.4 Kernel merging

Different learned profiles can produce correlated responses on training images. We use these correlations to propose groups of kernels for merging, then evaluate the resulting banks on validation data. Let $\mathbf{q}_j$ collect the response-map values of kernel j over a fixed subsample of training regions, and let ρ_{ij} be the Pearson correlation of $\mathbf{q}_i$ and $\mathbf{q}_j$. Define the dissimilarity

$$d_{ij} = 1 - |\rho_{ij}|. \tag{5}$$

Equation (5) treats a kernel and its sign reversal as equivalent candidates for merging. Their mean and maximum magnitudes are identical, and their signed extreme responses differ only in sign, which a logistic coefficient can absorb. For merely correlated maps, however, equivalence is not guaranteed: correlation removes offsets, and the nonlinear pooling operations need not preserve the relation between maps. Likewise, zero correlation does not by itself establish complementary diagnostic information.

Average-linkage hierarchical clustering starts with one cluster per kernel and repeatedly joins the two clusters with the smallest mean pairwise dissimilarity from Eq. (5). Cutting this hierarchy yields candidate partitions with K' clusters. For each partition, every cluster is replaced by

one profile; kernels are not removed individually within a cluster. The reference member is the one with the largest sum of absolute correlations to the other members. Profiles negatively correlated with this member are multiplied by -1, then the profiles are averaged and rescaled to their mean pre-merge norm. Expanding the average through Eq. (1) preserves the radial representation.

For every resulting bank, we recompute training-set feature standardization and refit the logistic classifier in Eq. (3). A candidate is eligible if its validation AUC and validation accuracy are each at least the corresponding full-bank value minus $\tau = 0.005$. Accuracy uses a probability threshold chosen on validation data. We evaluate the available cluster counts from 16 down to one and select the eligible bank with the fewest kernels; an ineligible intermediate count does not stop the search. We call this the *merged radial bank*. It is minimal only among the tested partitions under this criterion. Test scores are reported for comparison and are not part of the selection rule.

2.5 Comparison models and evaluation

Predefined-kernel bank. A predefined kernel has weights specified by a mathematical family and its parameters. Candidates are generated from nine families: Gaussian, anisotropic Gaussian, difference-of-Gaussians, Laplacian-of-Gaussian, Gabor, derivative, two-point co-occurrence, center–neighbor local-contrast, and discrete-Laplacian kernels. Parameters are drawn either independently at random or using Sobol quasi-Monte Carlo (QMC) sequences to distribute candidates over the parameter ranges. Each candidate is represented by the maximum magnitude $b_j(x)$ already defined in Eq. (2).

Candidates are ranked by training AUC, with the Fisher score (squared difference of class means divided by the sum of within-class variances) breaking ties. Maximal marginal relevance (MMR) [10] selects from this ranking while penalizing correlated region-level features. After selecting the highest-ranked candidate, each subsequent index is

$$j^* = \arg\max_{j \notin S} \left[\lambda A_j - (1-\lambda) \max_{\ell \in S} |\rho(\mathbf{b}_j, \mathbf{b}_\ell)| \right], \quad (6)$$

where S contains selected indices, A_j is the training AUC of candidate j, $\mathbf{b}_j = (b_j(x_i))_{i=1}^N$, and $\lambda = 0.75$. Unlike Eq. (5), Eq. (6) uses correlations between one pooled value per region, not between all response-map pixels. Both procedures discourage redundancy, but measure it at different stages. A logistic classifier is trained on the selected features. The reported predefined-kernel configurations use 31×31 kernels, up to 4,500 candidates, batch size 64, learning rate 5×10^{-4}, and 60 training epochs for the classifier.

Variants of the predefined-kernel bank. A *composite kernel* is a normalized weighted sum of selected kernels, $h_{\text{comp}} = \sum_i w_i h_i$, where w_i are the combination weights. Eight composites, or a single composite, replace the selected bank in the corresponding comparisons. A *rotation-averaged composite* averages this kernel over rotations: $h_{\text{rot}} = \frac{1}{4} \sum_{q=0}^{3} \text{rot}_{90q}(h_{\text{comp}})$, where rot_{90q} rotates the weights by $90q$ degrees. This enforces right-angle symmetry but need not produce a radial kernel. A second variant averages rotations every 10°. Radial-bin averaging is a separate step used to initialize the Breast learned radial bank (Section 2.3).

Adaptive best-subset selection (ABESS) [9] provides an alternative to MMR by fitting a classifier subject to a bound on the number of nonzero feature coefficients. A separate bank of triangular and square kernels tests predefined shapes outside the radial representation. These variants compare different representations and selection rules; they are not repeated estimates of the same model.

CNN comparison. For comparison with a multilayer learned representation, we use a CNN with four convolutional blocks, each including batch normalization, a rectified linear activation, max pooling, and dropout. Adaptive average pooling and a two-layer classifier follow these blocks. Input regions are 128×128 pixels with red, green, and blue channels. Training uses cross-entropy and AdamW with learning rate 3×10^{-4}, weight decay 10^{-4}, and at most 25 epochs. Training stops after six consecutive epochs without validation-AUC improvement; epoch six gave the highest validation AUC. The available experiments include this comparison only for Breast, so no CNN comparison is claimed for Brain. Because the kernel-based models use grayscale input, the Breast comparison also differs in available color information.

Evaluation protocol. The primary metric is the area under the receiver operating characteristic curve (AUC), which measures how well scores rank positive above negative regions across thresholds. Accuracy is the proportion of correctly classified regions at a specified threshold. Table 2 uses the default threshold 0.5. Tables 3 and 4 use a threshold selected to maximize validation accuracy. These conventions are stated separately because threshold choice can affect accuracy, particularly for Brain. Kernel selection, profile-checkpoint selection, and merging use training or validation data according to the procedures above. Test metrics describe the resulting models on the fixed test split. The archived predefined-kernel comparisons do not all use the same validation region set: several contain 1,790 regions, whereas the radial-bank experiments contain 1,870. The 90-kernel QMC configurations in Table 2 distinguish these validation counts. Some archived configurations also cap the training regions per class. These results are therefore descriptive comparisons of saved experiments, not a fully matched study of model architecture. No training-size sweep, repeated-split uncertainty estimate, or external validation is included.

3 Results

3.1 Breast classification at the available training size

Table 2 compares the recorded Breast classification results. The merged radial bank has test AUC 0.9415 with nine kernels, compared with 0.9411 for the learned radial bank with 16 kernels and 0.9387 for the QMC predefined-kernel bank with 90 kernels. The 0.0028 difference from the latter is small, comes from separately configured experiments, and has no uncertainty estimate; the clearer observation is that comparable discrimination is retained with one tenth as many kernels. The predefined-kernel bank has higher thresholded test accuracy, 0.8866 versus 0.8727. The CNN achieves the highest test AUC, 0.9626, and accuracy, 0.9021. These comparisons support a compact representation at the observed training size, but do not show which model would perform best with fewer cases.

3.2 Effect of kernel merging

On Breast, the procedure in Section 2.4 reduces 16 learned kernels to nine, a 43.75% reduction. Test AUC changes from 0.9411 to 0.9415 and validation-threshold test accuracy from 0.8749 to 0.8733 (Table 3). One cluster contains six original profiles, two clusters contain two profiles each, and six profiles remain in separate clusters. This result shows that some of the response patterns in the 16-kernel bank can be combined while retaining the observed discrimination.

On Brain, the selected nine-kernel merged radial bank has the same cluster-size distribution but loses 0.0289 test AUC relative to the full bank, from 0.7624 to 0.7335. It satisfies the validation criterion, so validation-based retention of performance does not guarantee retention on the test set. This finding limits the evidence for merging when data are scarce: the validation sample must itself be adequate to distinguish candidate banks.

Table 3 also reports single-kernel comparisons. The Breast comparison is a separately trained profile constrained to decrease with radius, with $R_{\max} = 20$ and one maximum-magnitude feature. It is distinct from the freely optimized single radial kernel in Table 2, which reaches 0.9286 test AUC. The Brain single-kernel comparison uses three response statistics and reaches 0.6903. These results show lower discrimination for the tested single-kernel configurations, but do not establish that every possible single-kernel model is inadequate.

3.3 Effect of kernel count

The merging comparison in Table 4 holds the three statistics in Eq. (2) and the logistic fitting procedure fixed while varying the number of merged kernels. It evaluates compression of an already trained bank; it does not measure the effect of training a smaller bank from the outset.

On Breast, merging to one kernel gives test AUC 0.9346, compared with 0.9415 at nine kernels. Thus, much of the discrimination remains after substantial compression. The one-kernel merged model also exceeds the 0.9212 AUC of the separately trained monotone single-kernel comparison. However, the models differ in profile constraints, initialization, optimization history, pooled features, and classifier fitting. Their difference cannot be attributed solely to using three statistics instead of one. A matched experiment with a fixed kernel and alternative pooling choices would be needed to isolate that effect.

On Brain, test AUC generally declines as the bank is reduced, reaching 0.6671 at one kernel. The decline is not monotonic: eight kernels give higher test AUC than nine. This is a descriptive observation, not a basis for changing the validation-selected bank. The selected training checkpoint is epoch one for Breast and epoch 33 for Brain. These observations indicate different optimization behavior; they do not establish that initialization alone explains the Breast result.

3.4 Effect of intensity standardization

Standardizing each Breast region while keeping the remaining configuration fixed reduces test AUC from 0.9411 to 0.8605 and accuracy at threshold 0.5 from 0.8754 to 0.7898. The model therefore uses information associated with the intensity scale in this dataset. This comparison does not determine whether that information reflects tissue properties, image acquisition, or both, and should not be generalized to a new cohort without validation.

3.5 Predefined-kernel comparisons

Table 5 reports predefined-kernel banks with one, three, or five selected kernels. The two one-kernel configurations reach test AUCs of 0.9157 and 0.9179; the three-kernel configurations reach 0.9101 and 0.9239. These results provide additional examples of compact classifiers, although performance varies with candidate selection and classifier fitting. Similar AUC values can coexist with different accuracy values because AUC measures ranking while accuracy uses a fixed threshold.

Table 6 compares candidate generation and kernel composition. For the reported 90-kernel banks, QMC sampling reaches 0.9387 test AUC compared with 0.9195 for random sampling. Replacing a bank by eight composites lowers test AUC by 0.0340–0.0453, and the single-composite comparison loses 0.0742. Weighted summation before pooling therefore did not preserve the discrimination obtained from separate kernel features in these configurations.

Averaging a composite over right-angle rotations gives identical reported metrics under 0°, 90°, 180°, and 270° rotations of the test regions. Relative to the corresponding non-averaged composite, test AUC decreases by 0.018 and accuracy by 0.014; averaging every 10° gives slightly lower values. ABESS-selected banks reach test AUC 0.9236 with 200 features and 0.9216–0.9225 with three to five features, but have lower accuracy than the corresponding MMR configurations in the reported comparisons. These observations

Table 2: Breast results. Val denotes validation; Acc. denotes accuracy at the default 0.5 threshold. "Kernels" is the number of retained kernels; the CNN learns its filters end to end.

Model / configuration	Kernels	Val AUC	Val Acc.	Test AUC	Test Acc.
Merged radial bank	**9**	**0.9373**	**0.8856**	**0.9415**	**0.8727**
Learned radial bank (16 kernels)	16	0.9388	0.8861	0.9411	0.8754
Single radial kernel (free profile)	1	0.9189	0.8545	0.9286	0.8668
CNN comparison	–	0.9445	0.8818	**0.9626**	**0.9021**
Predefined-kernel bank, QMC (1,790 val.)	90	0.9270	0.8603	0.9387	0.8866
Predefined-kernel bank, QMC (1,870 val.)	90	0.9271	0.8636	0.9377	0.8882
Predefined-kernel bank, random	90	0.9273	0.8620	0.9376	0.8888
Predefined-kernel bank (200 kernels)	200	0.9159	0.8514	0.9313	0.8578
Predefined-kernel bank (ABESS)	200	0.9095	0.7849	0.9236	0.7690
Triangular/square predefined-kernel bank	24	0.9109	0.8481	0.9221	0.8465

Table 3: Learned and merged radial banks, with separately trained single-kernel comparisons. "Features" is the number of scalars seen by the classifier. Merged banks are selected on validation data with $\tau = 0.005$; accuracy uses the validation-tuned threshold.

Cohort	Model	Kernels	Features	Val AUC	Test AUC	Test Acc.
Breast	Single radial kernel (monotone)	1	1	0.9100	0.9212	0.8412
	Learned radial bank	16	48	**0.9388**	0.9411	**0.8749**
	Merged radial bank	**9**	**27**	0.9373	**0.9415**	0.8733
Brain	Single radial kernel (free profile)	1	3	0.7281	0.6903	0.6365
	Learned radial bank	16	48	**0.7849**	**0.7624**	**0.6750**
	Merged radial bank	**9**	**27**	0.7801	0.7335	0.6640

Table 4: Kernel-count comparison. Each row refits a class-balanced logistic classifier on a bank merged to K' kernels. Asterisks mark validation-selected banks.

	Breast		Brain	
K'	Val AUC	Test AUC	Val AUC	Test AUC
16	0.9389	0.9411	0.7850	0.7624
13	0.9388	0.9412	0.7740	0.7593
11	0.9364	0.9407	0.7755	0.7531
9*	0.9373	0.9415	0.7801	0.7335
8	0.9365	0.9405	0.7794	0.7434
6	0.9299	0.9356	0.7593	0.7256
4	0.9295	0.9349	0.7220	0.7098
2	0.9259	0.9326	0.7039	0.6964
1	0.9271	0.9346	0.6218	0.6671

concern the evaluated configurations, not a general ranking of selection methods.

3.6 Interpreting a learned kernel

Figure 1 shows profile 14 of the saved 16-kernel Breast learned radial bank, using one-based numbering. Its weights are positive for radial bins 0–10 and negative for bins 11–21. The kernel therefore adds intensities in a central region and subtracts weighted intensities farther away. For example, increasing the central pixel intensity by ΔI, with other pixels fixed, changes the response at that position by $f_{14}(0)\Delta I \approx 0.425\Delta I$; increasing a pixel in bin 18 changes it by approximately $-0.183\Delta I$. This identifies the spatial comparison made by the kernel. Its weights do not sum to zero, so it also responds to overall intensity and is not a pure contrast measure.

The profile alone does not indicate whether a response favors the positive class. In the final logistic classifier for this 16-kernel bank, the coefficients for profile 14's standardized mean magnitude, maximum magnitude, and signed extreme response are -2.769, -1.727, and $+0.307$, respectively. By Eq. (3), a one-training-standard-deviation increase in mean magnitude, with all other features fixed, lowers the positive-class log odds by 2.769. Thus, this spatial pattern contributes through both the magnitude and polarity of its response, with distinct fitted effects. This is a conditional description of the classifier, not a causal statement about tissue.

Profile 14 remains a singleton during the nine-kernel merging step, so its spatial pattern is retained. The classifier is refitted after merging; the coefficients above belong to the 16-kernel model and must not be transferred to the merged model. Interpreting a retained profile requires its radial weights, response map, and refitted classifier coefficients. Assigning a tissue meaning also requires expert assessment.

Table 5: Compact MMR-selected predefined-kernel banks on Breast. Candidate counts are before selection; these archived experiments use 1,790 validation regions.

Compact configuration	Kernels	Val AUC	Val Acc.	Test AUC	Test Acc.
One predefined kernel, 9 candidates	1	0.8994	0.8307	0.9157	0.8449
One predefined kernel, 27 candidates	1	0.9010	0.8330	0.9179	0.8428
Three predefined kernels, 27 candidates	3	0.9085	0.8497	0.9101	0.8417
Three predefined kernels, 1,800 candidates	3	0.9085	0.8011	0.9239	0.7914
Five predefined kernels, 1,800 candidates	5	0.9071	0.7765	0.9234	0.7594

Table 6: Effect of candidate generation and composition on predefined-kernel banks with 90 selected kernels (Breast test set). Composite models replace the selected bank by eight composites unless noted.

Configuration	Base AUC	Base Acc.	Composite AUC	Composite Acc.
Random sampling, configuration 1	0.9195	0.8646	0.8855	0.8063
Sobol QMC sampling	0.9387	0.8866	0.8934	0.8176
Random sampling, configuration 2	0.9376	0.8888	0.9026	0.8299
Single-composite compression	0.9377	–	0.8635	0.7797

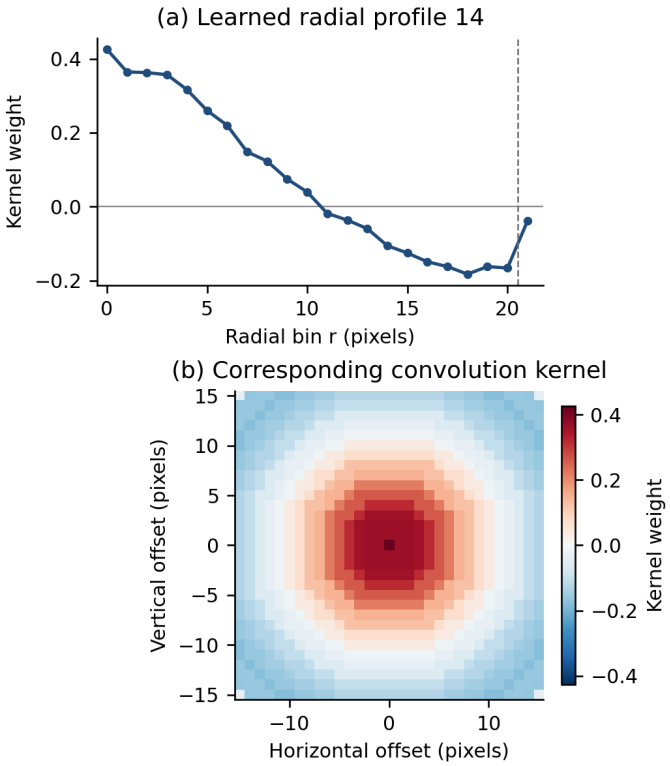


Figure 1: Profile 14 from the saved Breast learned radial bank: radial coefficients (top) and the corresponding 31×31 kernel (bottom). The vertical line separates optimized bins 0–20 from the fixed outer bin. Positive and negative weights identify the spatial intensity comparison; classifier coefficients determine its effect on the prediction.

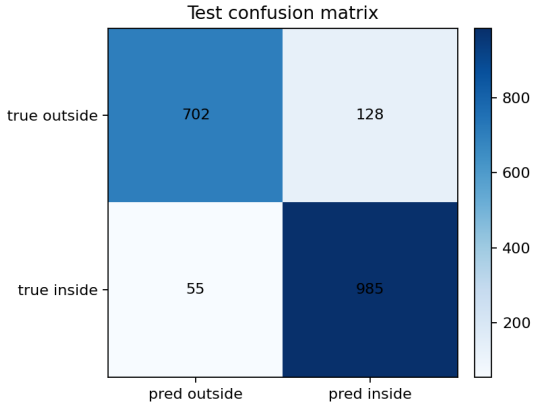


Figure 2: CNN comparison on the Breast test set of 1,870 regions. "Outside" and "inside" denote the negative and positive annotation-derived classes.

3.7 Qualitative classification maps

Figure 2 shows the CNN test confusion matrix; Figures 3 and 4 show example probability overlays. The overlays apply the merged radial classifier to a sliding grid of regions using the same pooled statistics as in training. They visualize region-level scores, not independent pixel-level predictions. No quantitative localization evaluation is reported here, and these examples do not establish which tissue structures account for a high score.

4 Discussion and Conclusion

The main contribution is a constrained classifier intended for settings where only a small number of annotated cases can be obtained. Radial parameterization limits each kernel

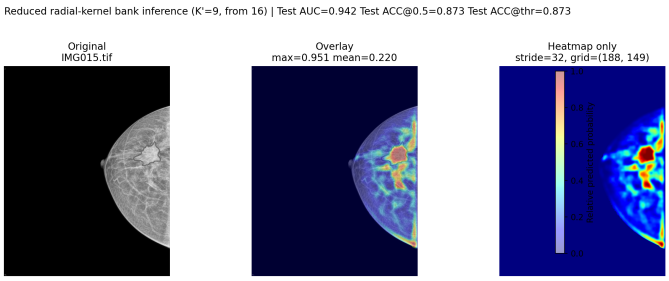


Figure 3: Breast example: region-level probability overlay from the merged radial bank.

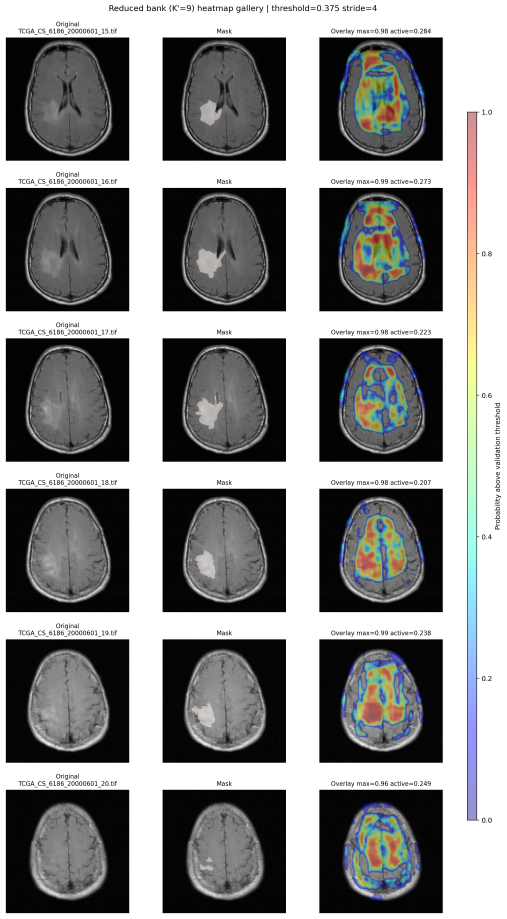


Figure 4: Brain examples: image, tumor mask, and merged radial bank probability overlay.

to 21 trainable coefficients for Breast or 31 for Brain, and the final classifier uses three explicit measurements per kernel. A second contribution is a validation-based merging procedure that reduces the number of kernels after training. On Breast, nine kernels retain the discrimination of the 16-kernel learned radial bank and perform comparably in AUC to a 90-kernel predefined-kernel bank. On Brain, the same reduction loses test AUC. These results establish a compact representation on the evaluated datasets, with dataset-dependent consequences of merging.

The intended benefit for small training sets remains a hypothesis rather than a demonstrated advantage. The radial-bank experiments use fixed training sets containing 6,330 Breast regions or 6,000 Brain regions; the archived comparison models have additional configuration differences described in Section 2.5. Multiple regions from a source image are dependent, so the region counts overstate the number of independent observations available for training. Moreover, reducing the deployed bank after fitting 16 profiles does not reduce the data used during that fitting. A direct test should train all compared models on nested subsets of source images or patients, repeat the subset selection, and keep validation and test groups separate. This would determine how performance changes as independent training cases become scarce and whether a smaller bank should be trained from the outset.

The merging results also show why model selection is consequential with limited data. Brain's selected nine-kernel bank meets the validation tolerance but loses 0.0289 test AUC. Sampling variability or differences between the validation and test groups could contribute, but the present experiment does not identify the cause. Repeated grouped splits could assess the stability of the selected bank size. The results do not justify choosing another size after inspecting its test score.

Interpretability is a secondary contribution. Section 3.6 demonstrates how a learned profile specifies a spatial intensity measurement and how logistic coefficients connect its pooled statistics to a prediction. This is more specific than labeling a small model interpretable solely because it has few kernels. It remains a mathematical account of the fitted classifier: a correspondence between a kernel and a named tissue structure has not been established. Correlated features also limit any interpretation of individual coefficients as unique measures of importance.

The representation comparisons provide additional qualifications. The merging sweep in Section 3.3 shows that Breast discrimination is retained with very few profiles, whereas Brain loses more discrimination under compression. It does not isolate the benefit of three pooled statistics over one, because the separately trained single-kernel models differ in other respects. Likewise, the Breast standardization comparison demonstrates sensitivity to intensity scale, but the experiments do not establish that standardization is necessary for Brain. The CNN has the highest Breast test AUC, although it receives color information absent from the kernel-based models. A matched grayscale comparison and a Brain CNN comparison are needed before attributing

differences to model architecture.

Further limitations include source-image rather than verified patient-level grouping for Breast, annotation-derived region labels, and the absence of repeated-seed or repeated-split confidence intervals. Small AUC differences should therefore not be treated as established improvements. Classification is evaluated at the region level; the illustrative maps do not establish localization performance or clinical utility, and neither dataset provides prospective external validation. The next empirical priority is a grouped training-size study, followed by validation on independent cohorts. The present evidence supports radial kernels as a compact candidate for learning with limited data, while leaving their data requirements to be measured directly.

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